

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows:** 3,900
- Columns:** 18
- Key Features:**
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data:** 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python. Initial exploration involved using `df.info()` to check structure and `describe()` for summary statistics:

metric	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Review Rating	Subs
count	3900.0	3900.0	3900	3900	3900	3900.0	3863.0	3900
unique	NaN	NaN	2	25	4	NaN	NaN	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	NaN	No
mean	1950.5	44.06	NaN	NaN	NaN	59.76	3.75	NaN
std	1125.9	15.20	NaN	NaN	NaN	23.68	0.71	NaN

Key Data Engineering Steps:

- Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- Column Standardization:** Renamed columns to snake case for better readability and documentation.
- Feature Engineering:** Created `age_group` column by binning customer ages, and `purchase_frequency_days` from existing purchase frequency data.
- Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender: Compared total revenue generated by male vs. female customers.

id	gender (text)	revenue (numeric)
1	Female	\$75,191
2	Male	\$157,890

2. High-Spending Discount Users: Identified customers who used discounts but still spent above the average purchase amount. *(Total rows: 839)*

id	customer_id (bigint)	purchase_amount (bigint)
1	2	\$64
2	3	\$73
3	4	\$90
4	7	\$85
5	9	\$97
...	...	...

3. Top 5 Products by Rating: Found products with the highest average review ratings.

id	item_purchased (text)	Average Product Rating (numeric)
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison: Compared average purchase amounts between Standard and Express shipping.

id	shipping_type (text)	Average (numeric)
1	Standard	\$58.46
2	Express	\$60.48

5. Subscribers vs. Non-Subscribers: Compared average spend and total revenue across subscription status.

id	subscription_status (text)	total_customers (bigint)	avg_spend (numeric)	total_revenue (numeric)
1	Yes	1053	\$59.49	\$62,645.00
2	No	2847	\$59.87	\$170,436.00

6. Discount-Dependent Products: Identified 5 products with the highest percentage of discounted purchases.

id	item_purchased (text)	discount_rate (%)
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer Segmentation: Classified customers into New, Returning, and Loyal segments based on purchase history.

id	customer_segment (text)	Number of Customers (bigint)
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 Products per Category: Listed the most purchased products within each category.

id	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions: Checked whether customers with >5 purchases are more likely to subscribe.

id	subscription_status (text)	repeat_buyers (bigint)
1	No	2518
2	Yes	958

10. Revenue by Age Group: Calculated total revenue contribution of each age group.

id	age_group (text)	total_revenue (numeric)
1	Young Adult	\$62,143
2	Middle-aged	\$59,197
3	Adult	\$55,978
4	Senior	\$55,763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present key insights visually, allowing stakeholders to filter and explore demographic and category trends dynamically.

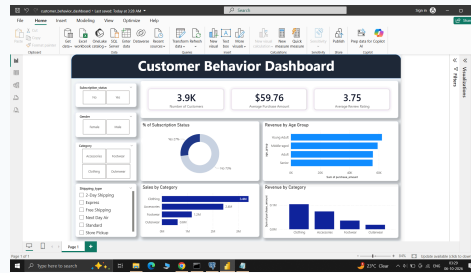


Figure 1: Interactive Customer Behavior Dashboard built in Power BI

6. Business Recommendations

- **Boost Subscriptions:** Promote exclusive benefits for subscribers to convert the large pool of repeat buyers (958 repeat buyers currently subscribed vs. 2518 unsubscribed).
- **Customer Loyalty Programs:** Reward repeat buyers to maintain their engagement and move them into the highly populated "Loyal" segment (3,116 customers).
- **Review Discount Policy:** Balance sales boosts with margin control, particularly for discount-dependent categories like Hats, Sneakers, and Coats which see ~50% discount usage rates.
- **Product Positioning:** Highlight top-rated and best-selling products (like Jewelry, Blouses, and Gloves) prominently in new marketing campaigns.
- **Targeted Marketing:** Focus efforts on high-revenue age groups (Young Adults and Middle-aged segments) and customers who utilize express shipping services, as they represent higher average order values.